

ULTRAHACK

Team 9 - RAPTOR

Explore – Detect – Respond – Fully Autonomous

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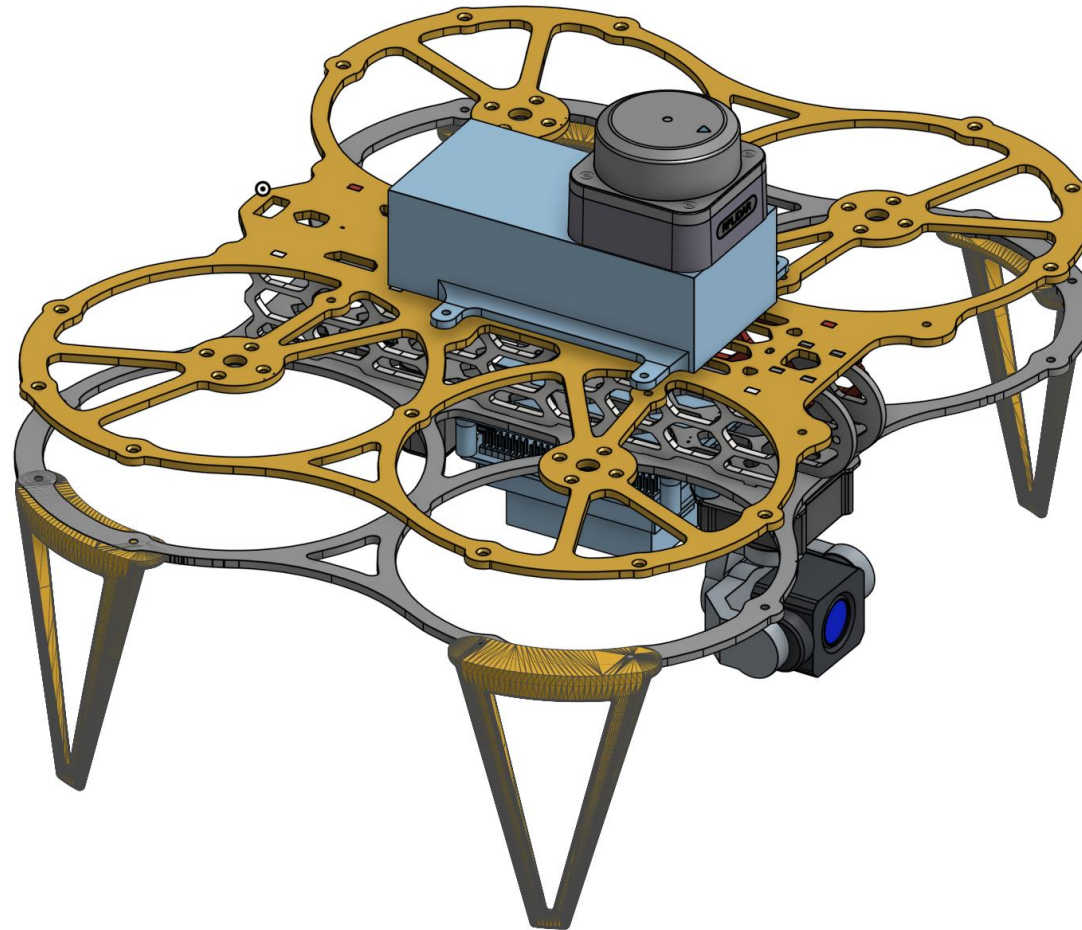


Introduction

RAPTOR stands for “**R**eal-time **A**utonomous **P**atrol for **T**tactical **O**peration & **R**econnaisance”

Autonomous Navigation

- LiDAR Path Planning
- Unmanned Deployment
- Rapid Mapping

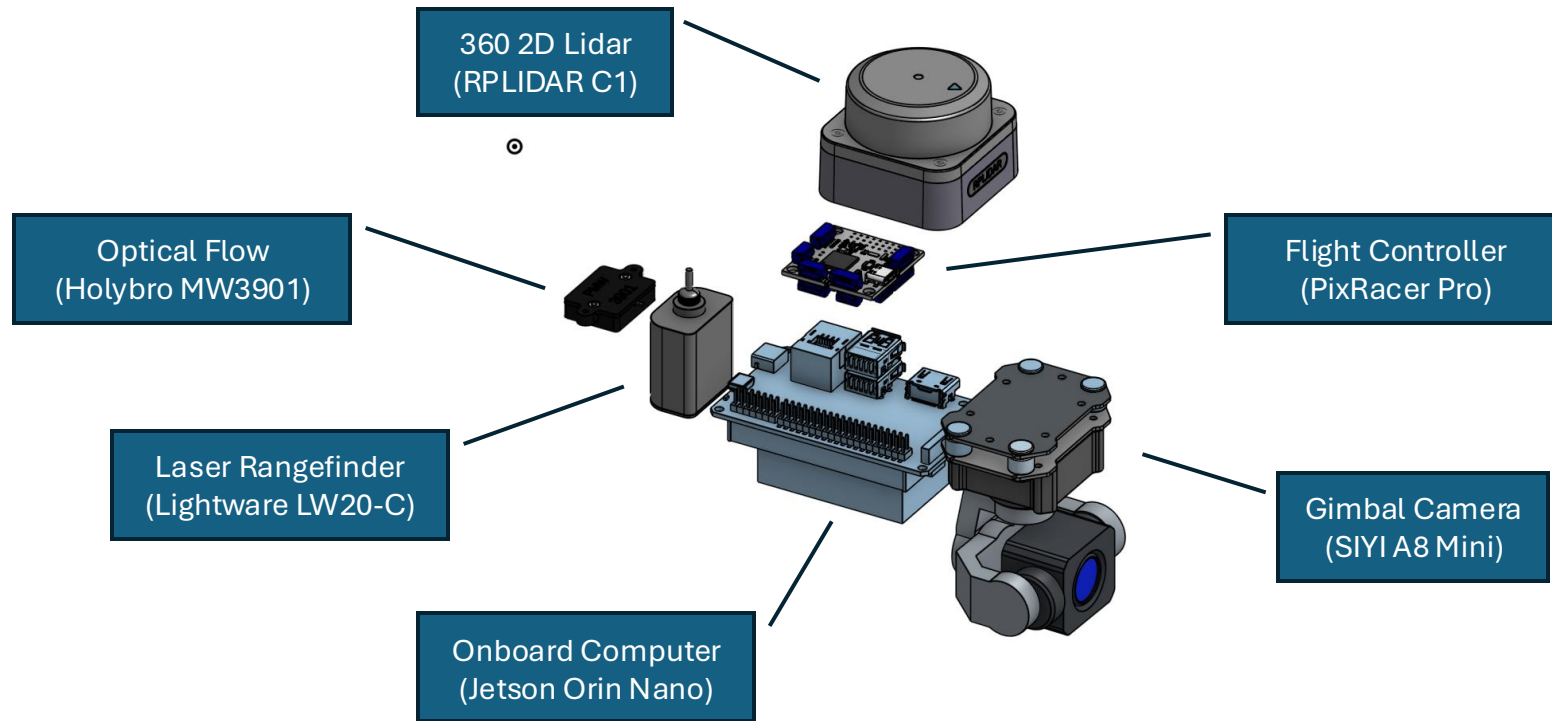


Edge Intelligence

- Real-time Detection
- On Device Processing
- Zero-Cloud Dependency

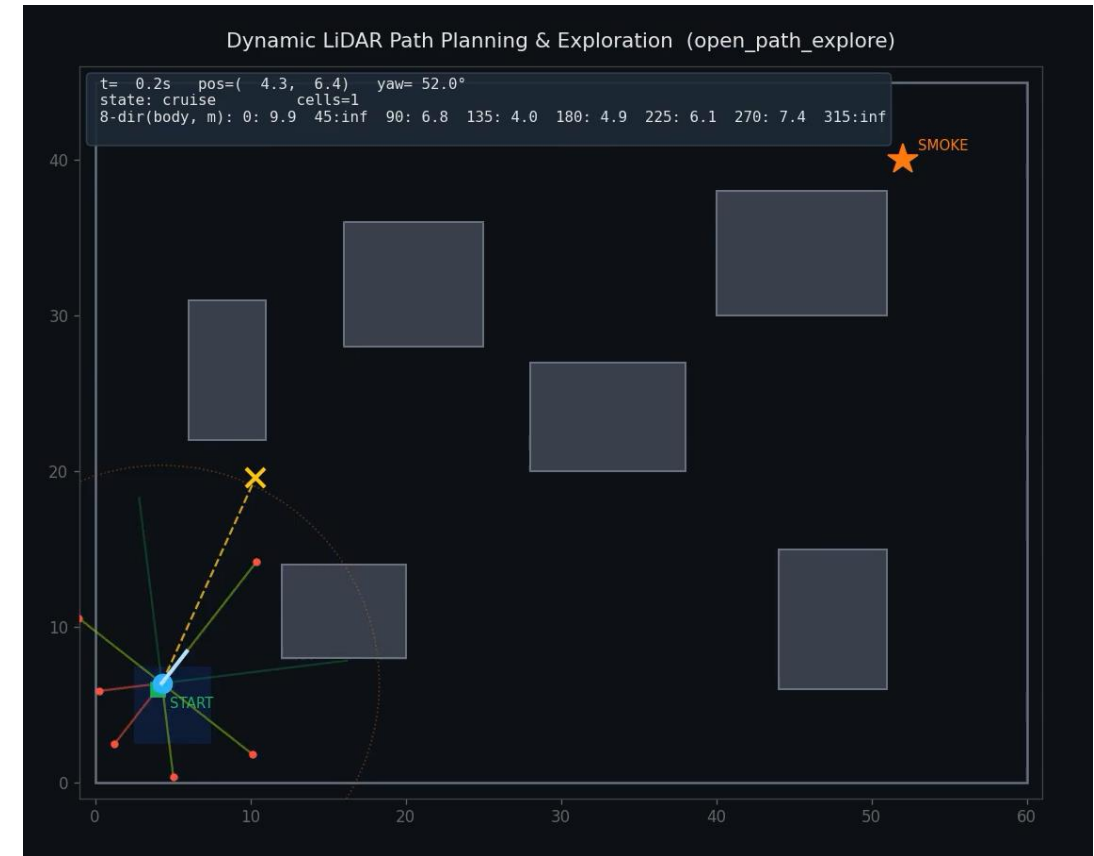
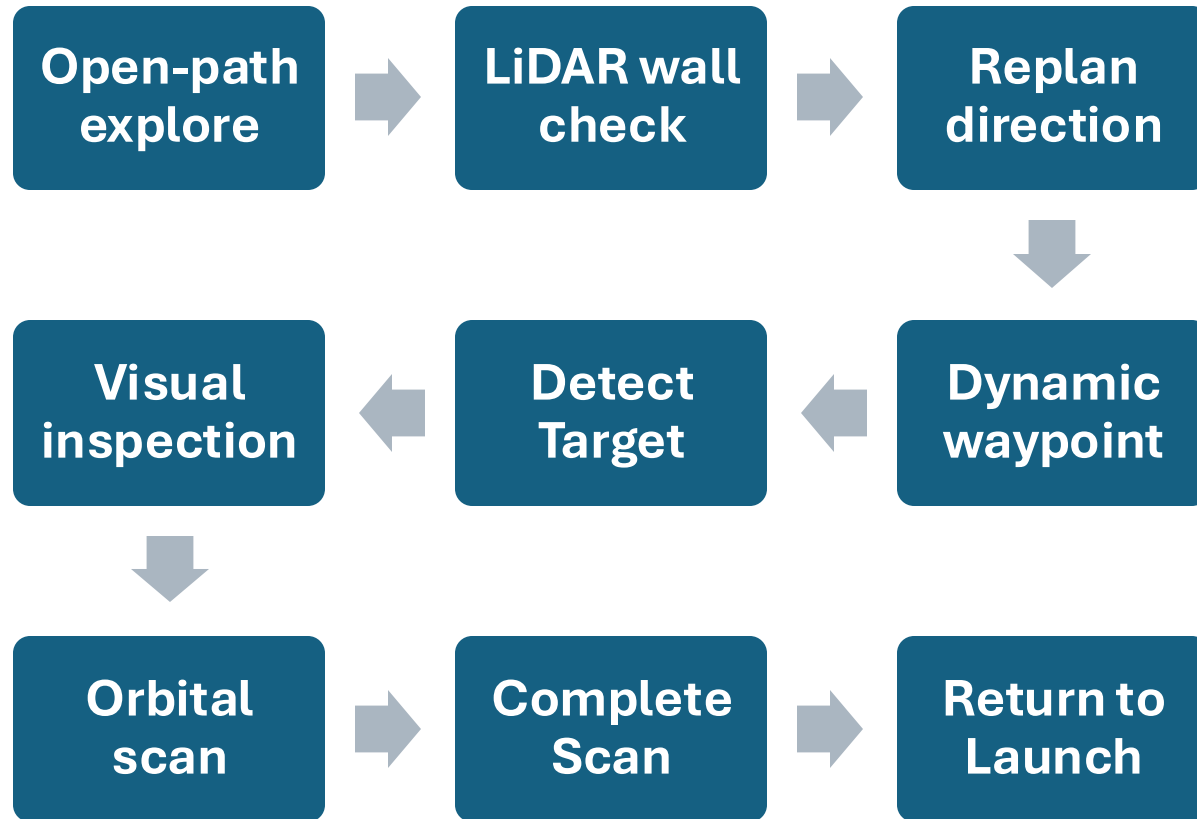
Components

RAPTOR is a self-contained autonomous survey stack, leveraging on board computing



Dynamic Path Planning

How RAPTOR covers the search area autonomously.



Real-time Detection

How RAPTOR identifies target with edge computing

Model: YOLOv26s

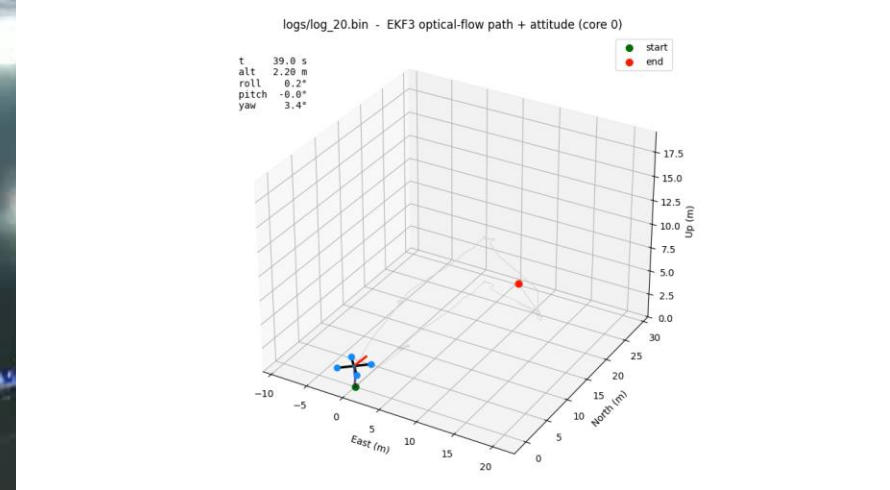
- Single-stage detector, classes: fire, smoke & human
- Runs on drone onboard computer
- TensorRT optimization with 16-bit quantization for low latency

Why On-Device?

- **Zero round-trip latency**: detection on the same frame the camera captures
- **Link-loss resilient**: keeps working even after the connection link is broken.

Dataset used for training & validation

Class	Train Images	Train Instances	Val Images	Val Instances
Fire	16,915	33,773	1,436	2,336
Smoke	28,769	32,538	6,735	7,090
Human	18,525	67,992	4,804	16,612
Lake	12,646	12,646	1,087	1,426
Total	40,384	146,949	11,953	27,464



Flight Execution

Goal: autonomous detection and surveillance mission in the Nokia Arena.

What worked

- Ran end-to-end in simulation, near-full in live flight
- YOLO detected the target, including real human
- Chose the right heading and closed in
- Camera recorded the target

What was less successful

- Imprecise return-to-launch (RTL drift)
- Marked a human behind smoke as critical point
- LIDAR missed the fence, drone collided with the net
- Didn't recognize artificial fire (trained on real outdoor fire/smoke)

Future Improvements

- **Indoor return-to-home:** GPS is inaccurate indoors; use visual-inertial odometry or VSLAM to navigate back to the launch point
- **Open-vocabulary detection:** replace fixed YOLO classes with an open-vocabulary model or VLM for richer, more reliable recognition
- **Robust obstacle avoidance:** add a depth camera to reliably detect thin obstacles like nets that LIDAR missed

Thank You

Team 9 - RAPTOR · github.com/akashshingha850/ultrahack2026